Appendix C - Down-Sampling Protocol

Regional Habitat Mapping Down-Sampling Protocol

August 22, 2022

1.0 Introduction

Funded by the Massachusetts Clean Energy Center (MassCEC), INSPIRE Environmental and the Northeast Regional Ocean Council (NROC) are working together to develop standard approaches for synthesizing, visualizing, and disseminating high-resolution geophysical and imagery data to advance benthic habitat mapping in the wind energy areas off the Northeastern United States (US).

Federal agencies and other management bodies require the synthesis and visualization of diverse datasets to accurately assess and mitigate the potential impact of human activities in ecologically and economically important ecosystems. As offshore wind development gains momentum in the Northeastern US, there is a substantial need to characterize and display the distribution of benthic habitats in wind farm lease areas and cable route corridors. Understanding benthic habitat distribution is necessary not only for engineering logistics but also for the important ecological value these environments provide, for example, to commercial fisheries.

This project will develop best practices for 1) integrating geophysical and benthic assessment data to map benthic habitats; and 2) making those habitat data available to federal and state regulators and stakeholders in a vetted and established forum, specifically the Northeast Ocean Data Portal.

The protocol detailed below outlines how geophysical data are prepared and standardized prior to their incorporation into regional composite data products. The requested geophysical data include multibeam bathymetry and backscatter. If data providers prefer to down-sample their own high-resolution data before sharing them to the project team, the following protocol should be followed to ensure consistency to the extent feasible.

2.0 Methods

Down-sampling is the process of averaging raster cell values over larger areas, effectively creating a coarser raster dataset with fewer cells and a smaller file size. Such procedures are vital to combining multiple datasets that span large spatial extents while preserving enough data resolution for regional assessment purposes. As such, the goal herein is to detail the process of reducing raster data file sizes, which promotes more efficient data processing routines and online portal usage, without greatly compromising the resolution of said data. Input raster data will be down-sampled to a resolution of 4 meters.

The Resample tool available in Esri's ArcMap and ArcPro converts the resolution of a raster dataset by interpolating values across new pixel cells. There are four techniques for resampling (down-sampling) spatial data in ArcGIS: nearest neighbor, majority, bilinear, and cubic. Bilinear interpolation was determined to be the best option for this project because it is the most commonly used interpolation technique for continuous data, i.e., raster datasets. Additional tests on sample data with varying cell sizes over a range of depth values indicated the bilinear approach produced down-sampled raster files with the best data quality overall (i.e., minimized overestimation, data range preservation) and required minimal computational effort. The Resample tool can be utilized in ArcMap Versions 10.6, 10.7, and 10.8, and in ArcPro Versions 2.7, 2.8, and 2.9 with basic, standard, and advanced licenses.

3.0 Protocol – ArcMap (Version 10.7.1)

3.1 Input Data and Setup

Input raster datasets should be delivered with depths (z-axis) as negative values and specific mention of the coordinate system and vertical units applied to each dataset. These datasets can be delivered in any commonly used, industry standard format such: *.TIF/TIFF; *.GRD; *.FLT. Please do not submit a rendered RGB or hill-shaded format.

1. File name and directory (i.e., folder) structure:

   (a) Files should be named using the following naming convention: "EntityCode_DataYear_DataTypeCode_Resolution"

For example: OR_21_BT_50cm

Where EntityCode = OR (Orsted), DataYear = 21 (2021), DataTypeCode = BT (Bathymetry), Resolution = 50cm

These project codes will be provided to developers or INSPIRE Environmental personnel will rename data upon receipt.

   (b) Folders should be organized as below (Figure C-1):

- One main folder: Project_fromEntity

- One subfolder for the source bathymetry data: Project_DataYear_Bathy_Resolution
- One subfolder for the source backscatter data: Project _DataYear_BS_Resolution
- One subfolder for the down-sampled bathymetry raster: Project _DataYear_Bathy_Resolution
- One subfolder for the down-sampled backscatter raster: Project _DataYear_BS_Resolution

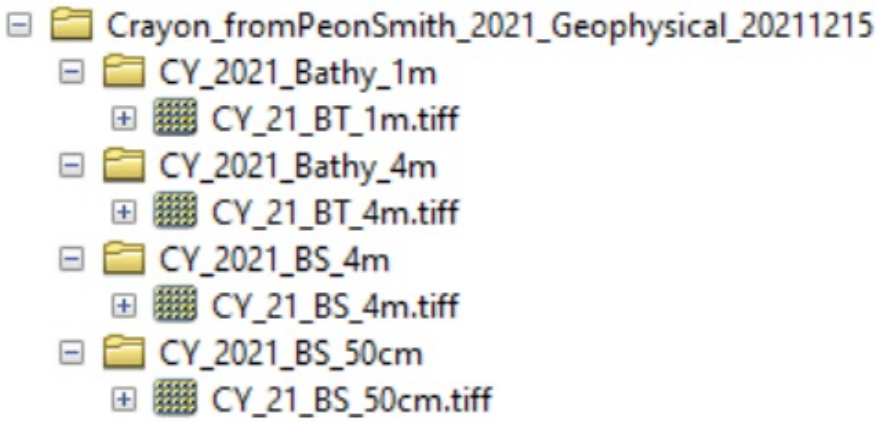


Figure C-1. Preferred folder structure (completed in ArcCatalog or File Explorer).

2. Map document and geoprocessing properties:

   (a) Open a new, blank ArcMap .MXD file

   (b) Leave all geoprocessing environment settings as default

3.2 Procedure

1. Adding raster files to the map:

   (a) Add the original raster files to the map using either the Add Data tool or the Catalog Tree.

   (b) Symbolize the raster files using the below properties:

Bathymetry: Minimum-Maximum stretch type, with the Invert box checked as below. These settings are within the Symbology tab of the Layer Properties (Figure C-2A).

Backscatter: Standard Deviation stretch type (n = 2). These settings are within the Symbology tab of the Layer Properties (Figure C-2B).

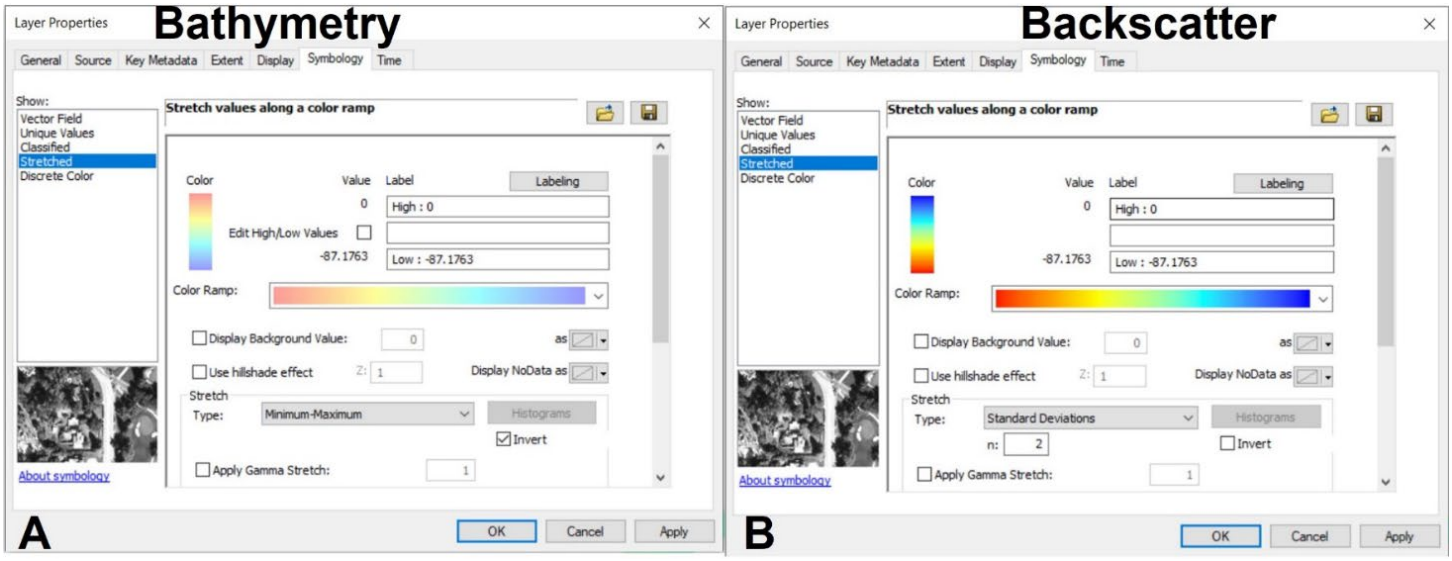


Figure C-2. Preferred symbology for (A) bathymetry and (B) backscatter raster file datasets.

2. Resample

(a) Search for the Resample (Data Management) tool using the search function in ArcMap (Figure C-3).

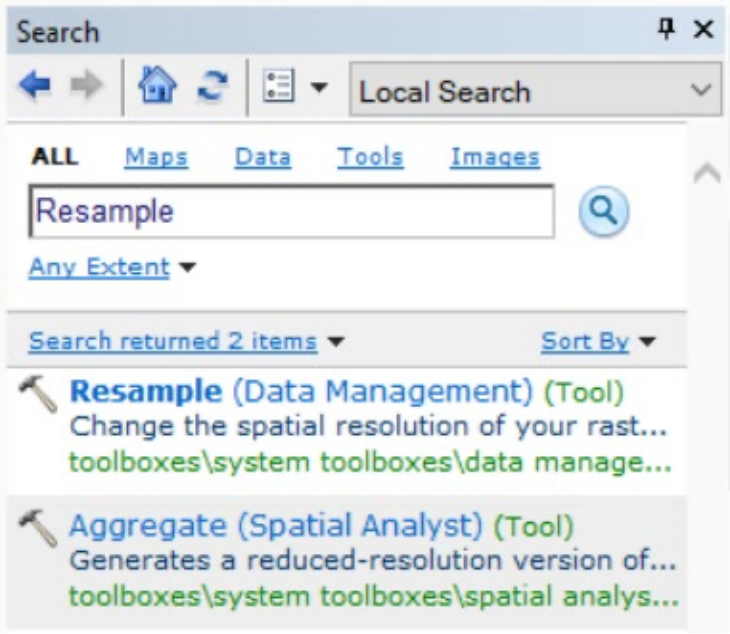


Figure C-3. The Resample tool can be found using the Search function in ArcMap.

(b) Select the Resample tool to open the tool's dialog box.

(c) Fill out the tool's parameter fields using the guidance below and Figure C-4.

i. Input Raster: the original data, stored in the folder created within Step 1(b) of Section 3.1. If the raster was added to the .MXD file, it can also be selected by using the dropdown menu (blue circle, Figure 4).

ii. Output Raster Dataset: location and name of the desired down-sampled raster. Use the folder icon (green circle, Figure 4) to navigate to the folder created in Step 1(b) of Section 3.1.

Note: *The output raster's name can only be 13 characters long. Therefore, it is essential that the file is named appropriately. The tool will signal if the name is too long with a red X icon by the Output Raster Dataset line.*

iii. Cell Size: enter the desired cell size (e.g., 8) for X and Y.

iv. Resampling Technique: the default is set to Nearest Neighbor. Change to Bilinear.

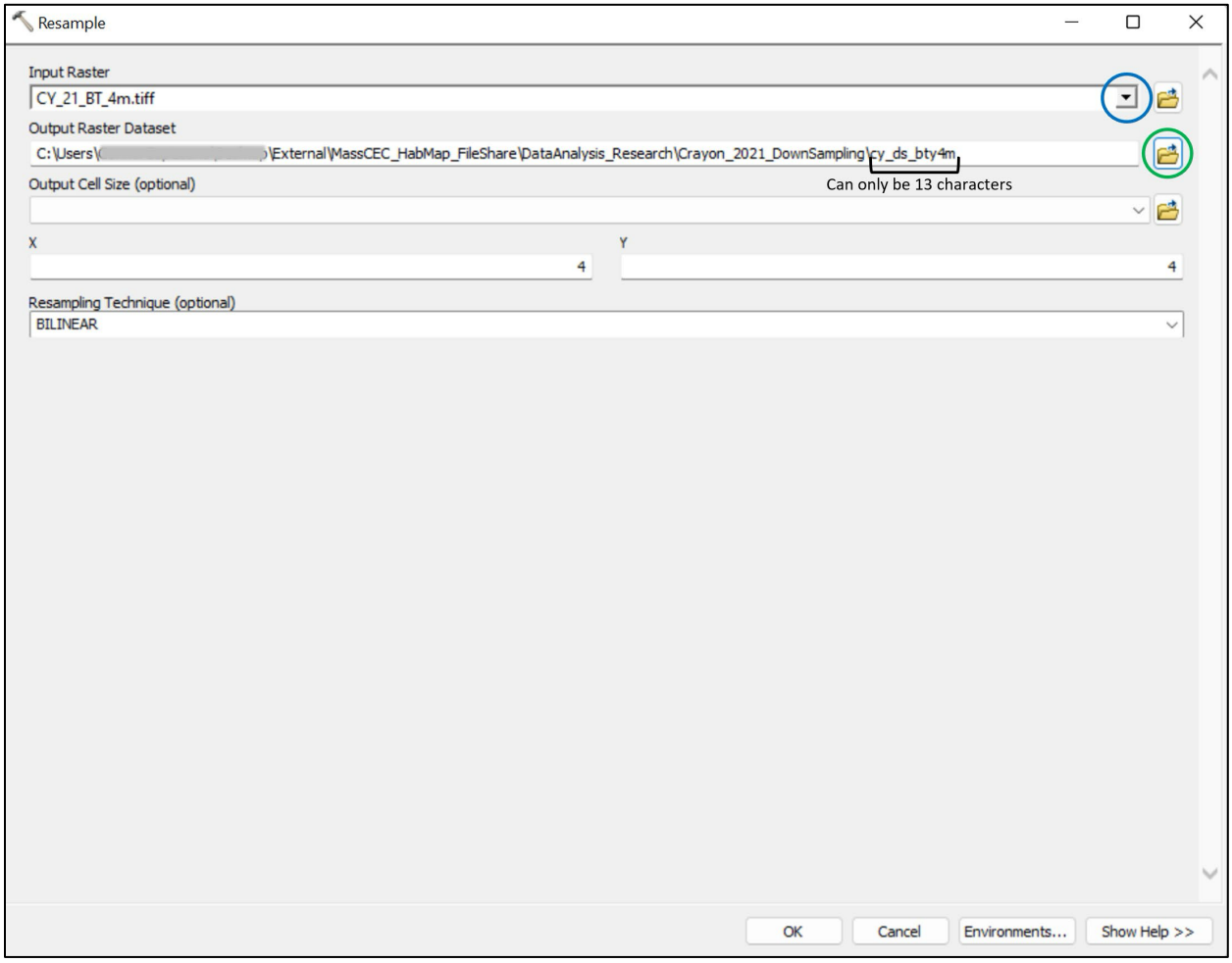


Figure C-4. The appropriate parameters and settings for the Resample tool.

(d) Symbolize the down-sampled raster using the same symbology identified in Step 1 of Section 3.2.

(e) Repeat the process for each bathymetry and backscatter raster file dataset as needed.

3. Creating a down-sampled hill-shaded relief model raster.

   (a) To create a hill-shaded raster (i.e., hill-shaded relief model) from bathymetry data, follow these steps:

      i. Under the Windows tab, select Image Analysis (Figure C-5).

      ii. Select the down-sampled bathymetry raster in the Image Analysis side panel (Figure C-6, blue rectangle).

      iii. Under the Display section, change the interpolation method to Bilinear Interpolation (Figure C-6, red rectangle).

      iv. Under the Processing section within the panel, select a black to white gradient color ramp (Figure C-6, green rectangle).

      v. Select the Shaded Relief icon next to the color ramp to create a hill-shaded relief model (Figure C-6, purple circle).

      vi. Symbolize the hill-shaded relief model using the symbology show in Figure C-7. When the Stretch Type is changed to Minimum-Maximum, allow ArcMap to calculate statistics if prompted.

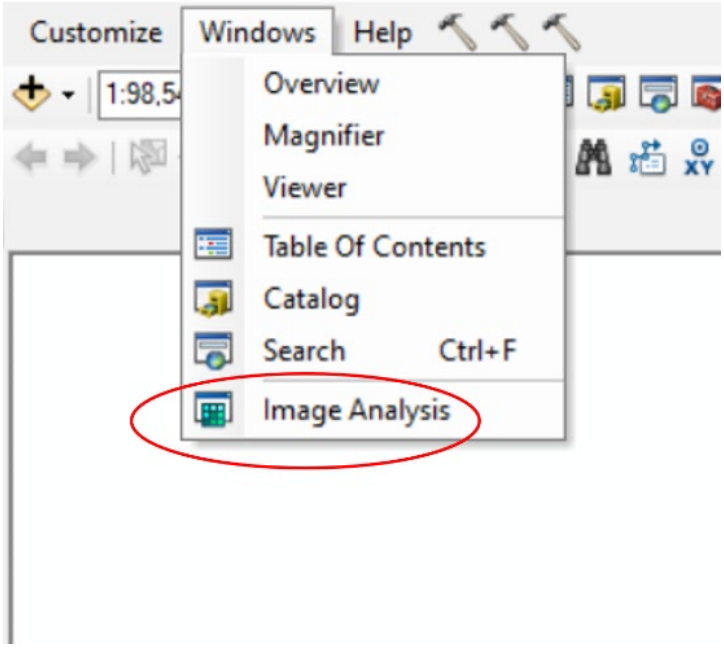


Figure C-5. The Image Analysis tool can be found under the Windows tab.

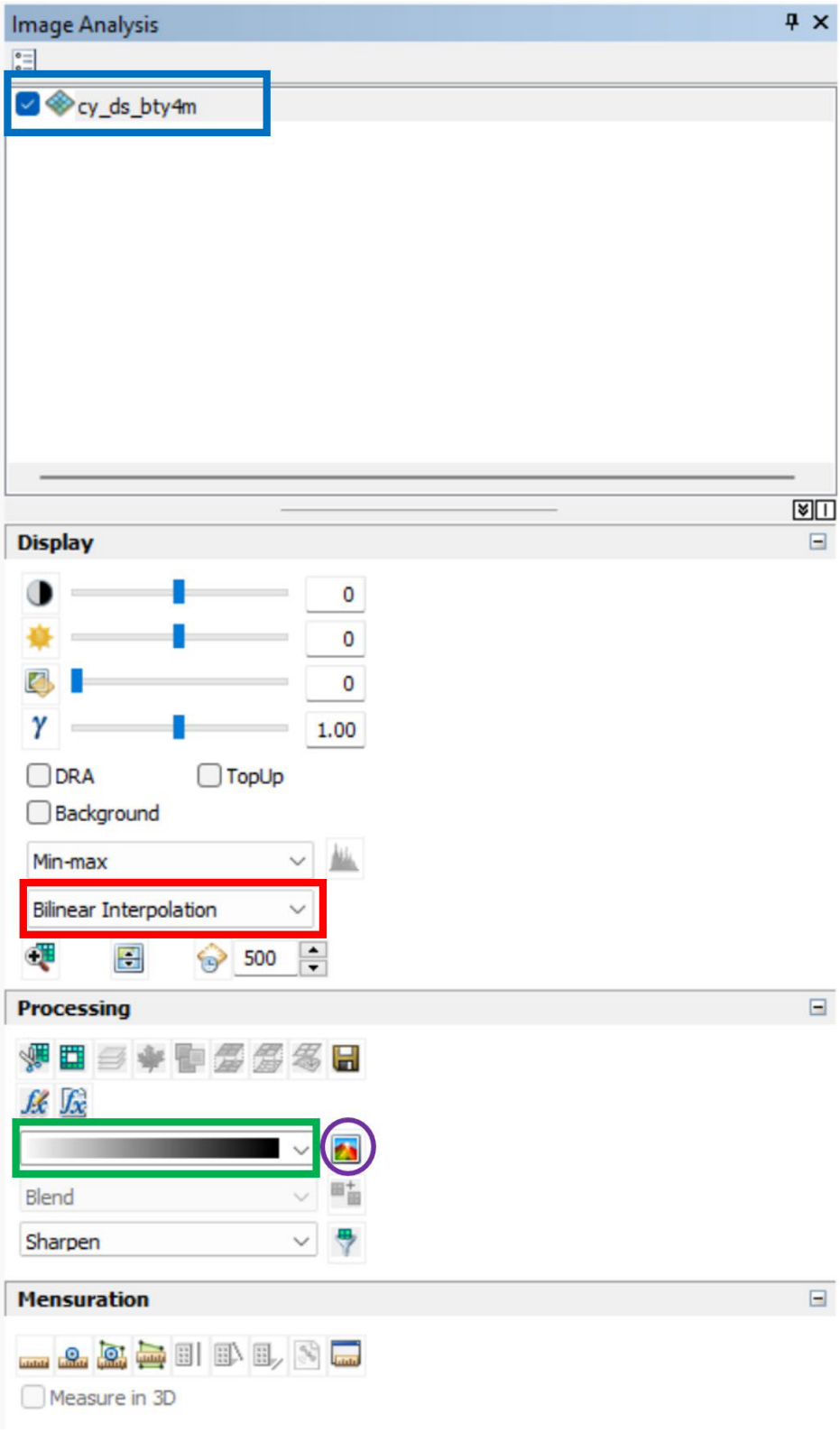


Figure C-6. Parameters within the Image Analysis tool to create a hill-shade relief model.

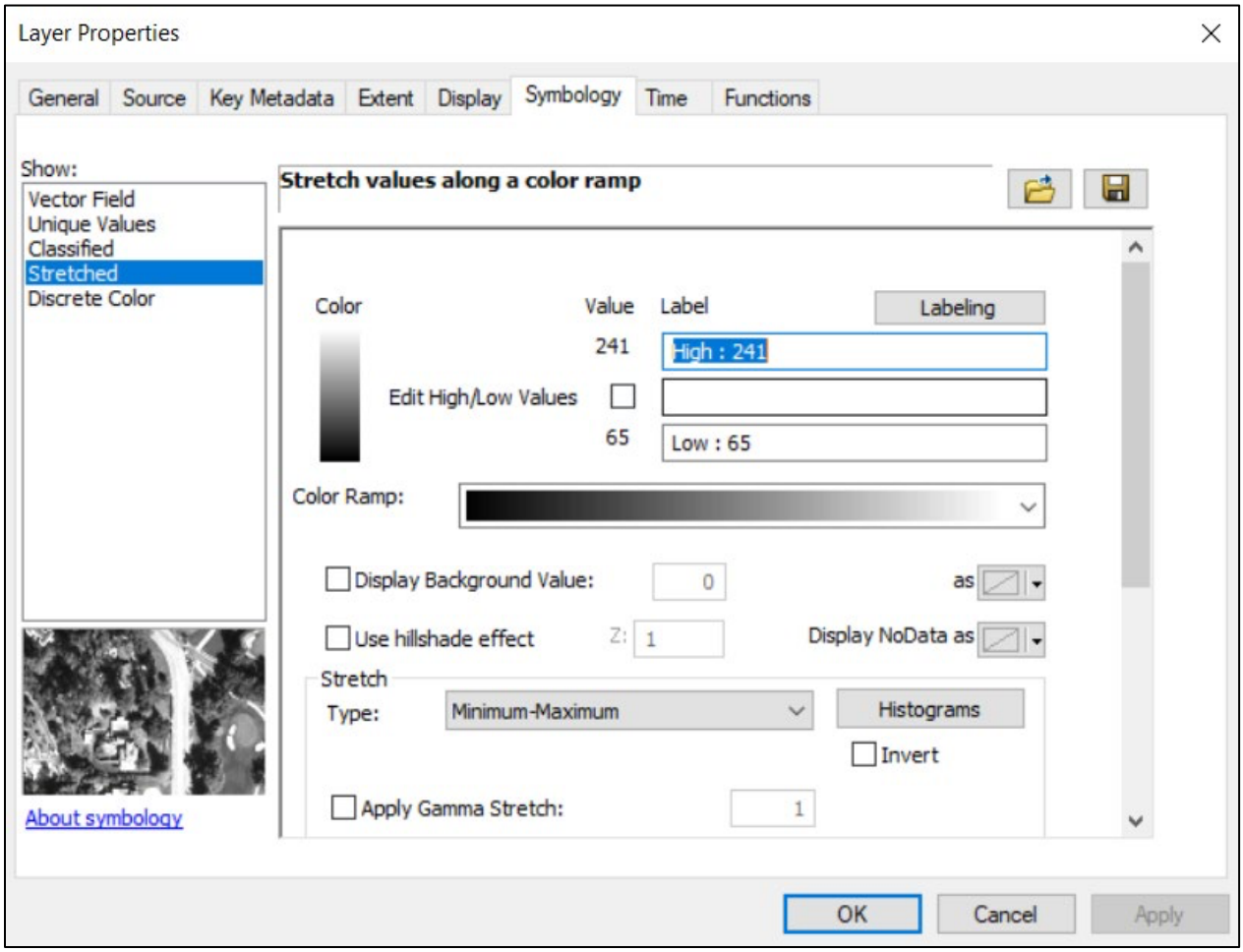


Figure C-7. The preferred symbology for bathymetry hill-shaded relief models.

4. Exporting Data

   (a) Down-sampled raster files are already saved within the folders created in Step 1(b) of Section 3.1.

   (b) Created hill-shaded relief models will need to be exported because they originally exist as temporary files (Figure C-8). This can be confirmed in the Source tab of the hill-shaded relief model's layer properties.

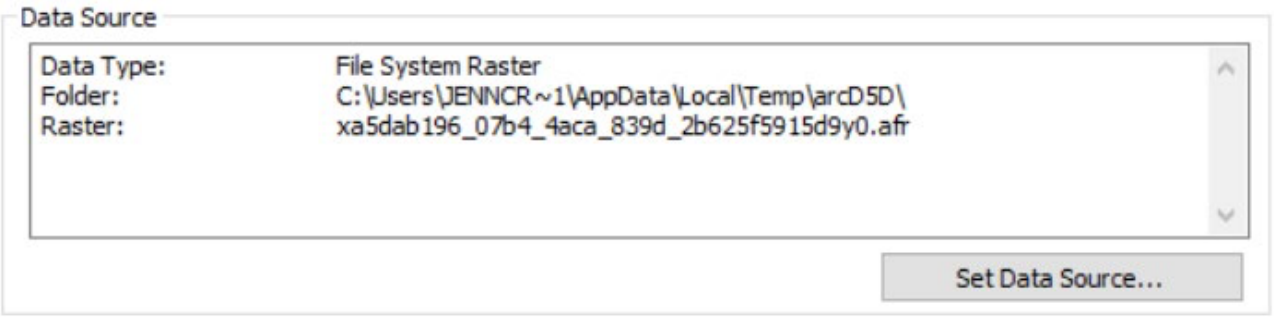


Figure C-8. Layers created in the Image Analysis tool are stored in a temporary area by default.

   (c) To export a hill-shaded relief model, right click on the layer name in the Table of Contents and select Data, then select Export Data.

   (d) Navigate to the folder with the down-sampled bathymetry raster file and set that folder as the Location (Figure C-9).

   (e) Name the hill-shaded relief model (remember, the name has a limit of 13 characters).

   (f) Select *.TIFF as a Format.

   (g) Save the file.

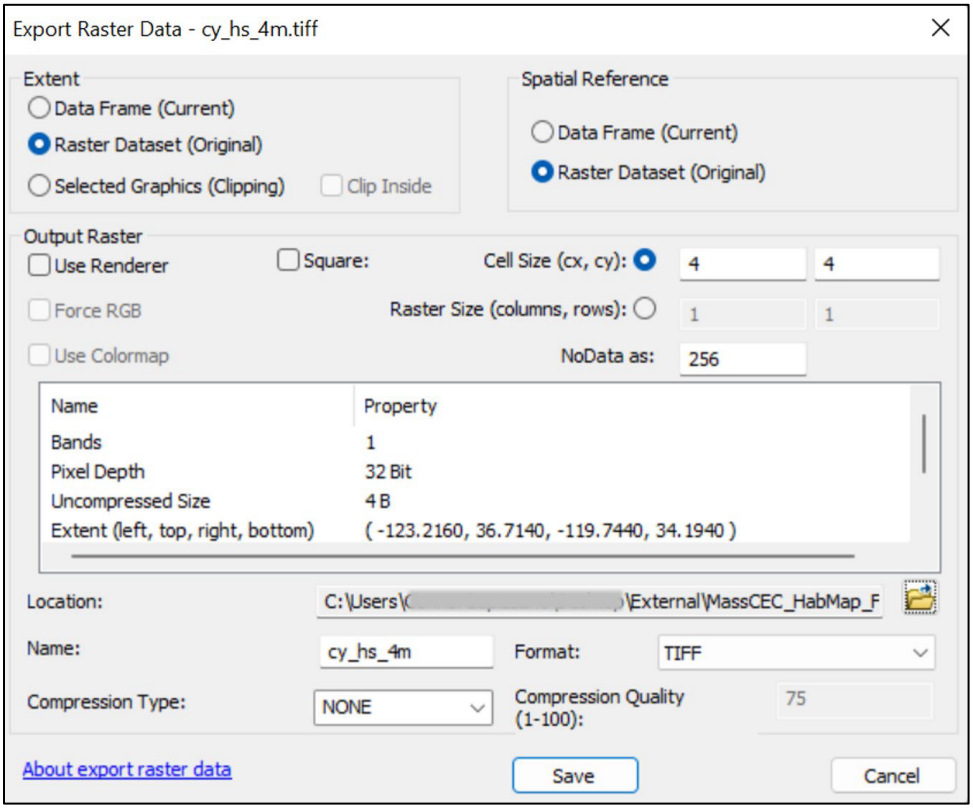


Figure C-9. The settings to properly export a temporary raster as a *.TIFF.